

Antaral

अंतराल • the interval

A digital twin for the gap between dialysis sessions

Team Between Beats

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Digital Twin Challenge 2026 • Happiest Health

The danger lives in the gap between sessions

~30%

of patients at one Bengaluru dialysis centre were on twice-weekly dialysis, with higher weight gains and lower dialysis adequacy [3]

Silent

Potassium builds up between sessions. Too much can trigger a fatal arrhythmia, often with no warning symptoms

Day after

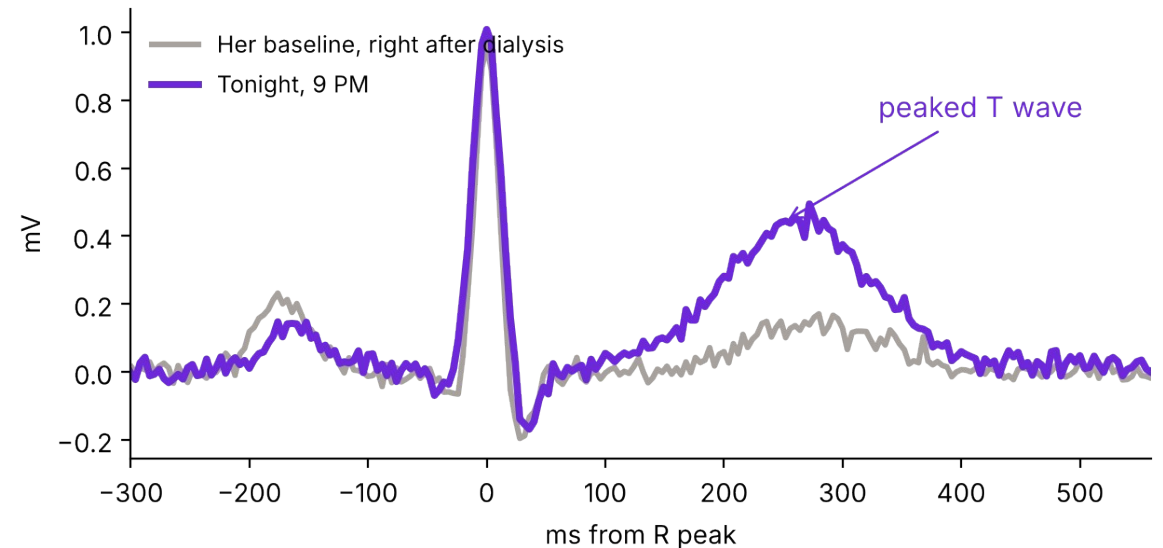
the long gap: deaths and cardiovascular hospitalisations cluster after the longest interval between sessions [1]

Blind

Clinics check potassium on a monthly lab, and every patient keeps fixed weekdays whatever their potassium is doing

Tuesday, 9 PM. Is she safe until Thursday?

- 58, diabetic, on an ARB. Dialysis Mon & Thu.
- Drank tender coconut water on Monday and Tuesday, and logged it.
- Tonight her smartwatch T wave is far taller than her post-dialysis baseline.
- Her next potassium lab is weeks away.

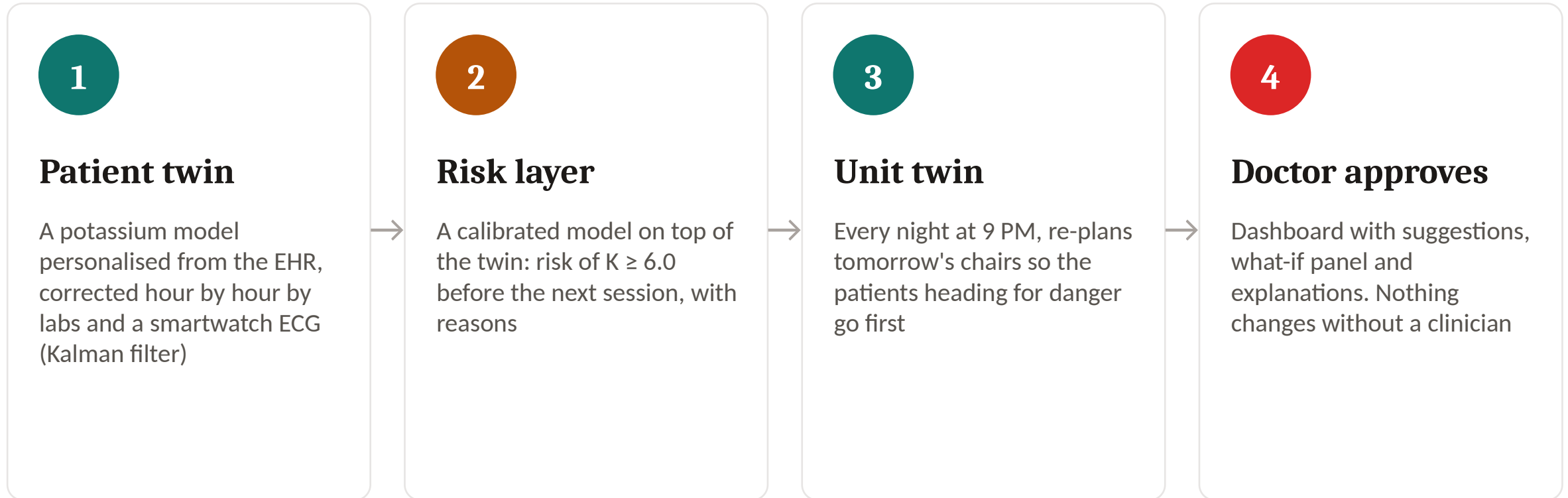


Smartwatch median beat: her baseline vs tonight

The nephrologist's question tonight

Should she come in tomorrow, or wait until Thursday?

Three twins, one evening decision



Same machines. Same number of sessions per patient. Smarter timing, earlier warnings.

Static record + live signals, fused hourly

Static EHR

- Age, sex, dry weight
- Diagnosis: diabetic nephropathy, CKD of unknown aetiology...
- ACE-i / ARB, beta-blocker
- Residual urine output, bicarbonate
- Dialysis pattern: 2× or 3× a week

Wearables & home

- Smartwatch single-lead ECG, twice a day
- Smart-scale morning weight
- Steps, resting HR, HRV, sleep, BP
- Food diary: coconut water, banana, dry fruits, aloo sabzi

Dialysis unit

- Session times and length
- Pre-dialysis weight
- Potassium lab, about monthly
- Chairs and shifts available

Sandbox rules: no real patient data. Antaral's own simulator generates the EHRs, the hidden physiology, the ECG waveforms and the wearables. Next: validation on MIMIC-IV.

A Kalman filter around real potassium physiology

Model

$dK/dt = (\text{absorbed intake} - \text{renal} + \text{tissue release}) / V$

Hidden state

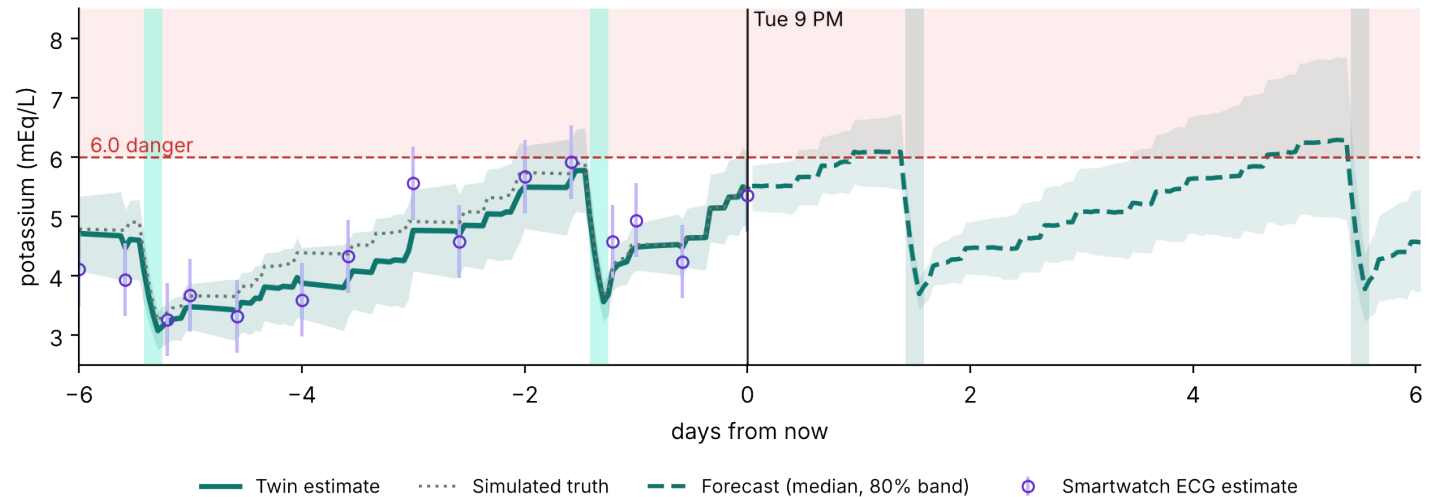
potassium K and a personal drift b , learned over days

Corrected by

monthly labs (precise) and smartwatch-ECG estimates (noisy)

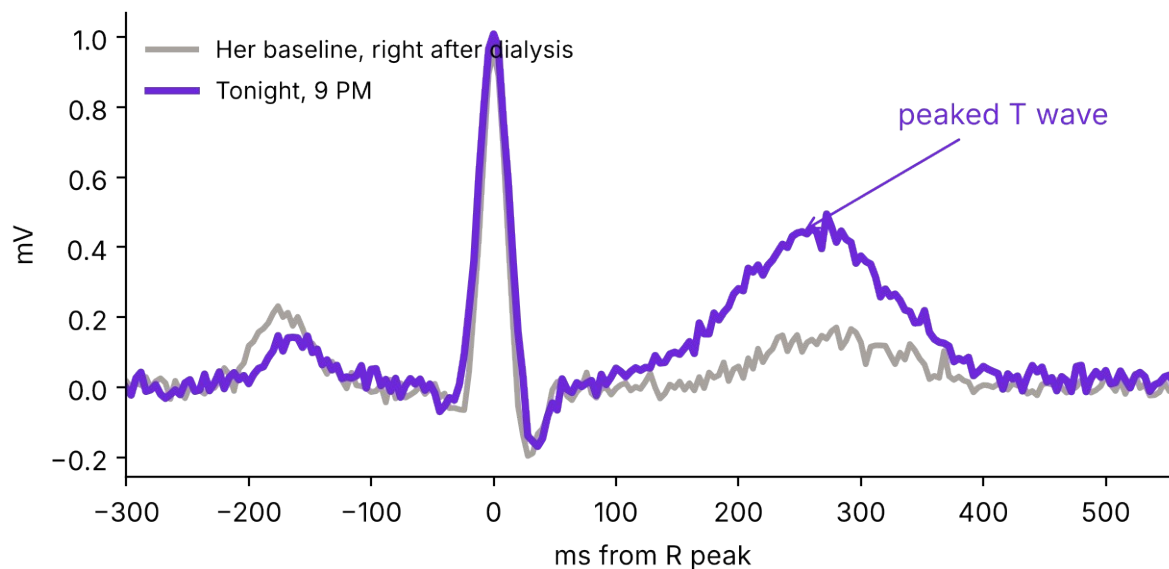
What-if

Monte Carlo rollouts for any session time, food, binder or diet call



Lakshmi's twin: estimate tracks the simulated truth (dotted); forecast crosses 6.0 before Thursday and again before Monday

Reading potassium from a peaked T wave



0.50

mEq/L mean absolute error on held-out patients

0.82

correlation with true potassium

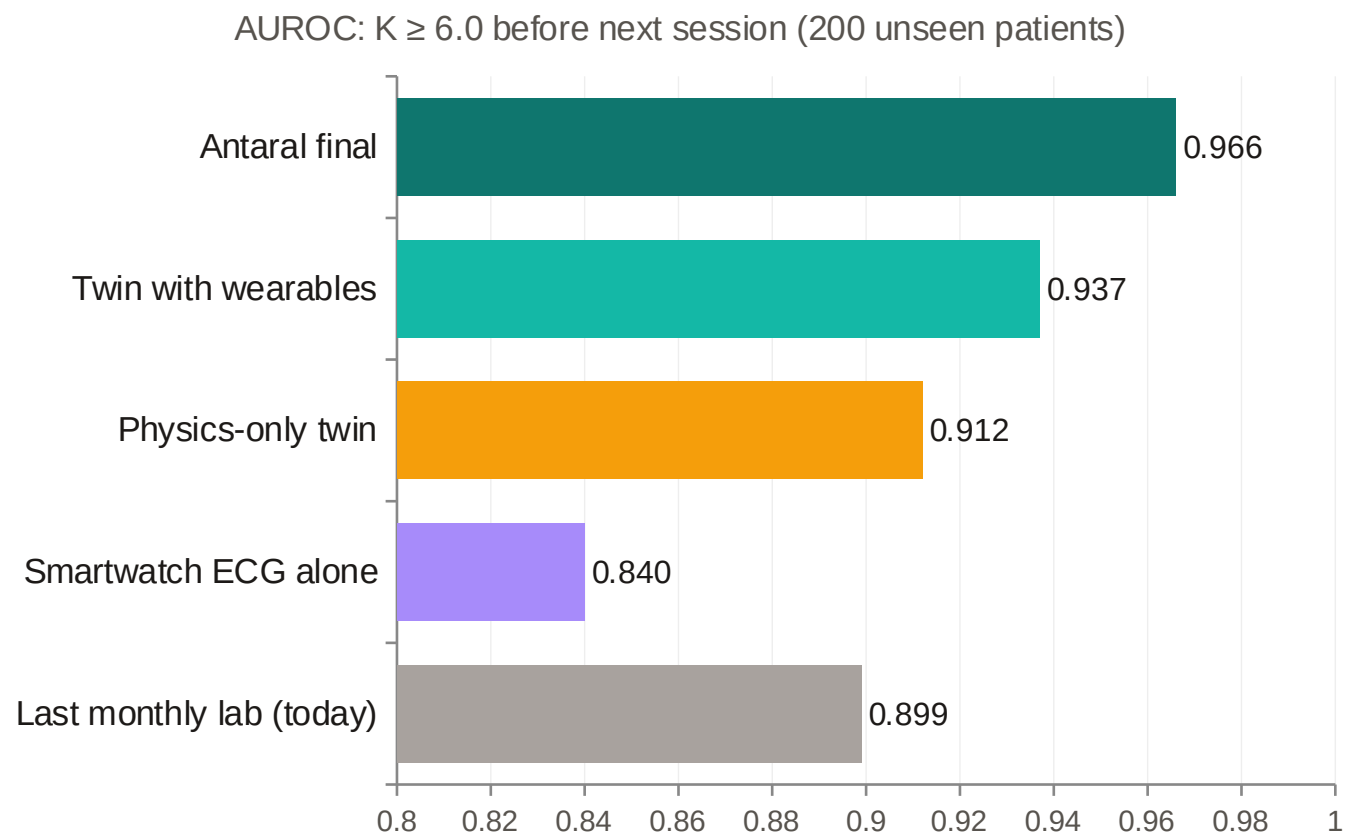
0.70 → 0.62

error (RMSE) after personalising to the patient's own baseline

High potassium: taller, narrower T wave; longer PR; wider QRS. Each beat is compared with the patient's own post-dialysis beat.

Deliberately noisy, like a real watch: the twin trusts it less than a lab. Deep learning on 2-lead ECGs has detected hyperkalaemia with AUROC 0.85–0.88 in large CKD cohorts [2].

Fusion beats every single data stream



89%

of dangerous nights caught at
90% specificity
vs 70% with the last monthly lab

0.85

within-patient AUROC: spots
her dangerous weeks
vs 0.56 for the lab

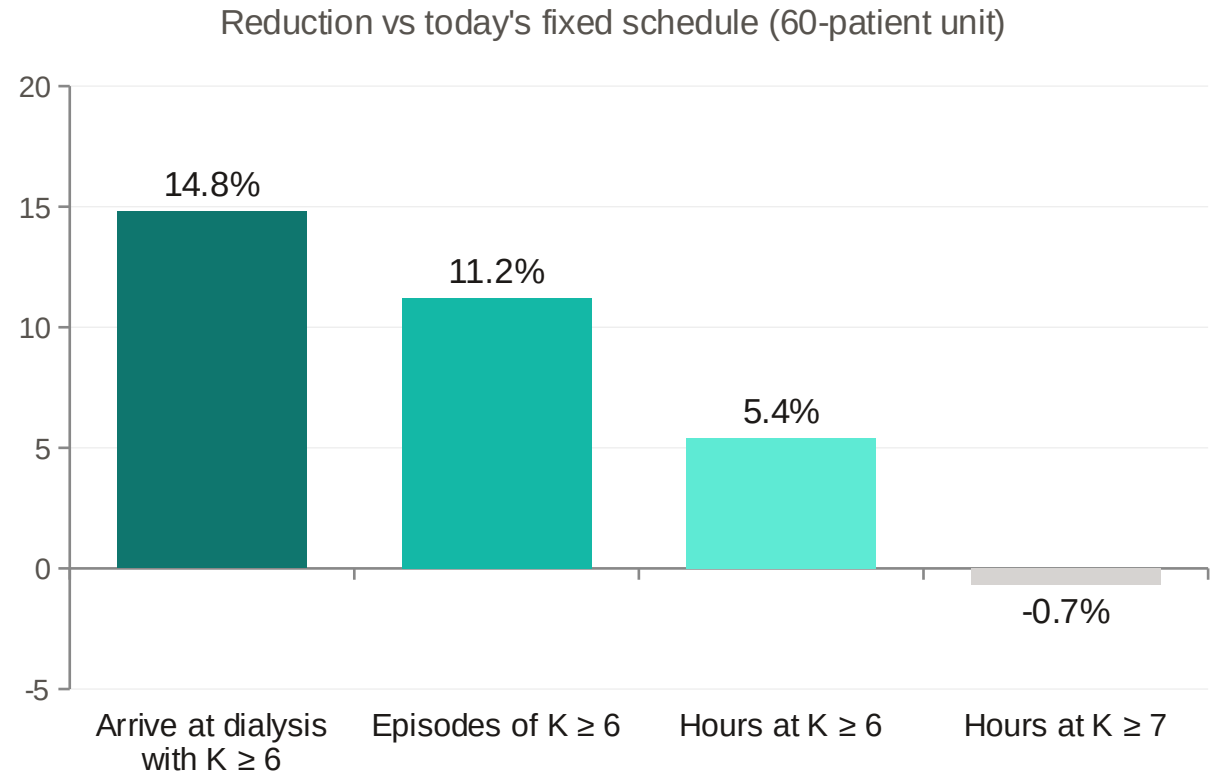
13,000

patient-nights on held-out
patients
95% CI for AUROC: 0.955–0.975

Re-timing chairs, not adding sessions

Every night at 9 PM an integer programme picks tomorrow's chairs:

- Maximise danger-hours avoided ($K \geq 6$, weighted up at 6.5 and 7.0)
- Same chairs, same sessions per patient, safe minimum and maximum gaps
- Looks at both upcoming gaps: moving a session earlier makes the next gap longer
- Riskiest patients get the earliest shift



6 replicate units $\times$ 8 weeks, identical patients, meals, illnesses and session counts. Fewer arrivals with $K \geq 6$ in all 6 units.

An AI that knows when NOT to move a patient

Keep Thursday

48

expected danger-hours over her next two gaps

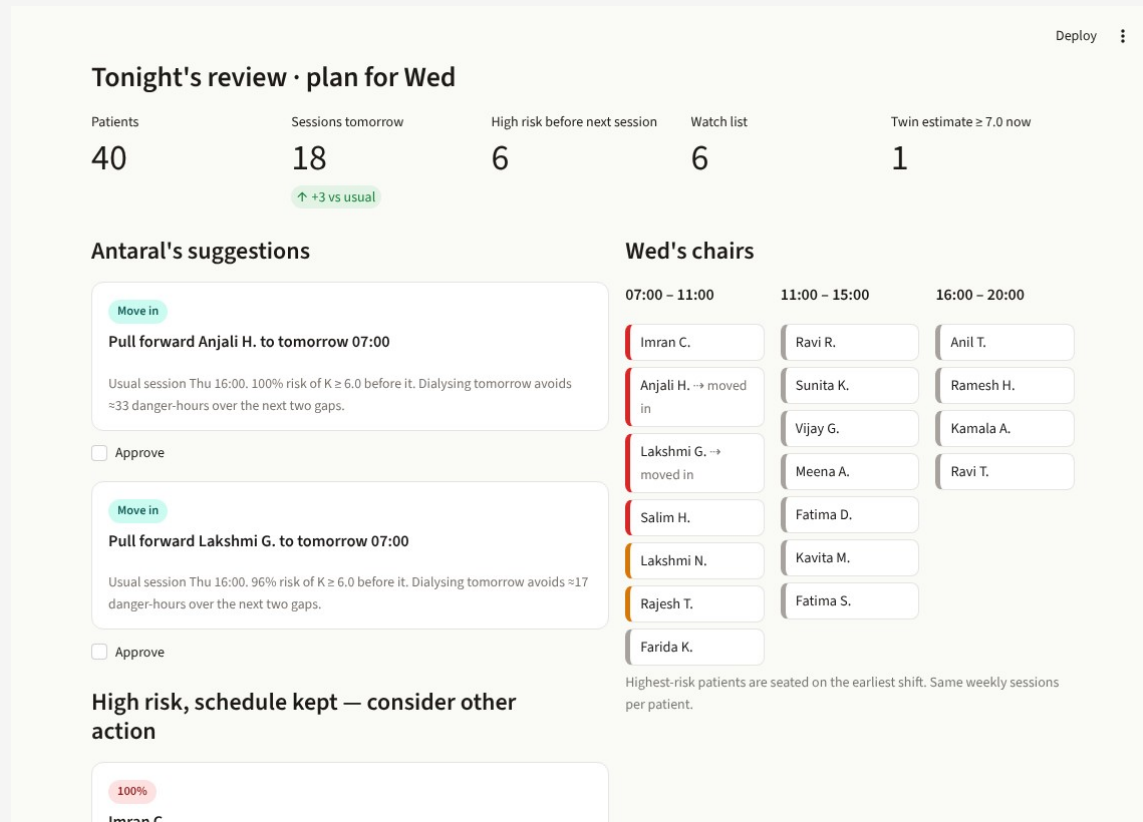
Move to Wednesday

55

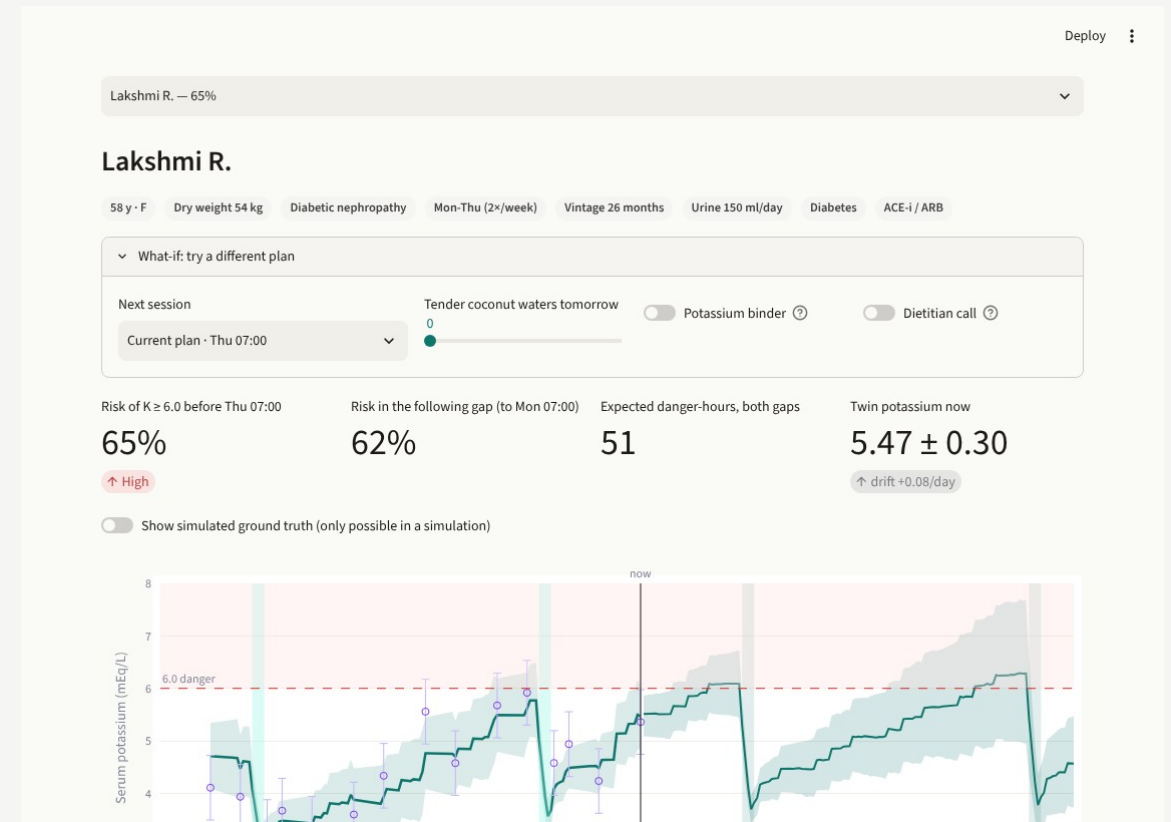
danger-hours: her next gap would stretch to 5 days, until Monday

So Antarl keeps her on Thursday, uses Wednesday's spare chairs for two patients it can actually help, and flags Lakshmi under "**high risk, schedule kept: consider other action**". In the what-if panel the doctor compares a potassium binder or a dietitian call about the coconut water.

How a doctor works with the virtual patient



Unit tonight: suggestions to approve, tomorrow's chair board, risk for every patient



Patient twin: trajectory, forecast, ECG vs baseline, reasons, what-if panel

What the simulation taught us, and what comes next

Limits we found

- Re-timing does not cut hours above 7.0: those patients need more dialysis, not different timing
- Too many schedule changes for a real unit today; the stability setting needs tuning with clinicians
- All results are on synthetic data, not clinical validation
- A tuned ML model without the twin comes close on accuracy; the twin's edge is what-if reasoning

Next

- Validate on MIMIC-IV: dialysis patients with potassium labs and linked ECGs
- Real smartwatch ECG exports in place of synthetic beats
- Dose alerts for chronically high patients: extra session or prescription review
- Pilot with an Indian dialysis unit, DPDP-compliant, clinician approval on every change



Same chairs. Same sessions. Fewer people arriving in danger.

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github.com/SillyMarvellousChad/BetweenBeats_VIT-Bhopal-University

Research prototype for clinician decision support. Synthetic data only. Not a medical device.

